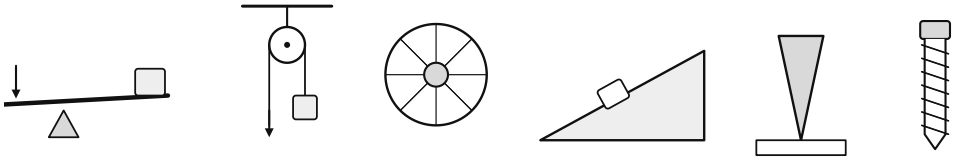
PHYSICAL AND CHEMICAL SCIENCE

Mechanical Systems

Forces, the six simple machines and how they combine in mechanical systems



In this pack

Learn it: Simple machines and mechanical systems ★☆☆
Fact check: Six simple machines ★☆☆
Think it through: Machines at work ★★☆
Investigate: How does distance from the pivot change the effort? ★★☆
Real-world case study: The rickshaw: a city on three wheels ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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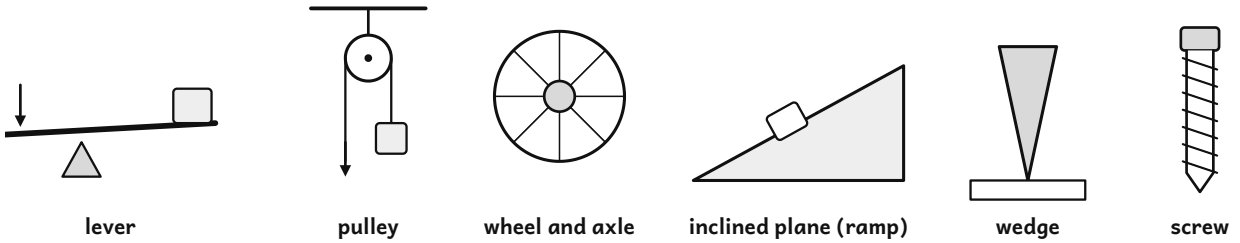
Simple machines and mechanical systems

Name: Class: Date:

Learning goals

- Name the six simple machines and explain how they make work easier.
- Identify simple machines inside mechanical systems.
- Explain how forces start and stop moving devices.

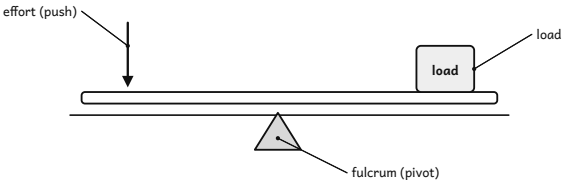
How does a small person lift a heavy bucket of water from a well, crack a hard nut, or carry three passengers through Dhaka on a rickshaw? They use **machines**. Machines help us by changing the **size** or **direction** of a **force** (a push or a pull).



The six simple machines.

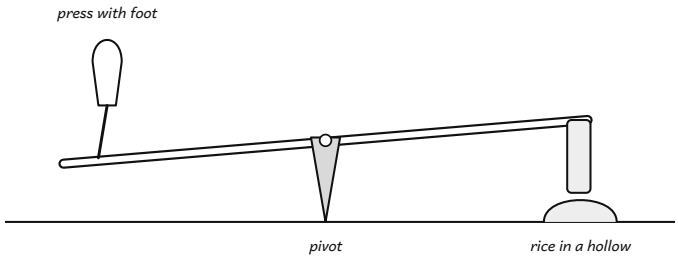
1. Lever

A **lever** is a bar that turns on a **pivot** (also called a **fulcrum**). You push on one part (the **effort**) to move a **load**. The further the effort is from the pivot, the **less force** you need. Examples: a see-saw, scissors, a bottle opener, a nutcracker, the handle of a tube-well, and the *dheki*.



Bangladesh connection

The **dheki** is a traditional Bengali rice-husker. A long wooden beam rests on a pivot. A person presses one end with their foot, and the heavy head at the other end lifts and then drops onto the rice, removing the husks.



A dheki is a lever.

2. Pulley

A **pulley** is a wheel with a rope around it. It changes the **direction** of a force: you pull **down** to lift a load **up**. Using several pulleys together means less force is needed. Examples: a flagpole, a well bucket, a crane.

3. Wheel and axle

A large **wheel** fixed to a thin rod (the **axle**) turns together with it. A small force on the big wheel makes a bigger turning force on the axle. Examples: rickshaw wheels, a door knob, a screwdriver, a steering wheel.

4. Inclined plane (ramp)

A **ramp** lets you raise a heavy load by pushing it a **longer** distance with **less** force. Examples: a wheelchair ramp, a sloping road up a bridge, a plank for loading a truck.

5. Wedge

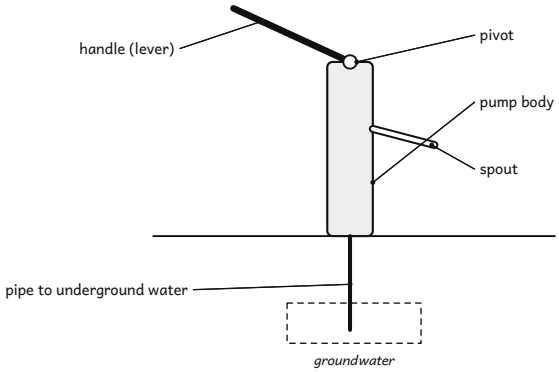
A **wedge** is two ramps back to back. It turns a push into a sideways force that **splits** or **cuts**. Examples: an axe, a knife, a nail point and the Bengali **boti**.

6. Screw

A **screw** is a ramp wrapped around a rod. Turning it moves it in or out with great force. Examples: jar lids, bolts, a bottle cap, a car jack.

Mechanical systems

A **mechanical system** is a device in which several simple machines work together. A bicycle or rickshaw uses pedals and cranks (levers and wheel and axle), a chain and gears, wheels and axles, and brake levers. Forces **start** it (the rider pushing the pedals), and forces **stop** it (**friction** from the brakes).



A hand tube-well is a mechanical system with a lever handle.

Key words

force	a push or a pull
lever	a bar that turns on a pivot
fulcrum (pivot)	the point a lever turns on
effort	the force you put in
load	the object being moved
pulley	a wheel with a rope that changes the direction of a force
friction	a force that slows moving things
mechanical system	simple machines working together

FACT CHECK · C6 · MECHANICAL SYSTEMS

Six simple machines

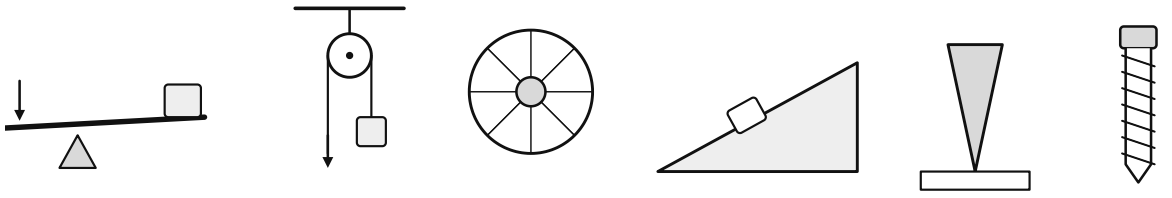
★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Name and identify simple machines.
- Label a lever.

1 Write the letter of each simple machine. [6]

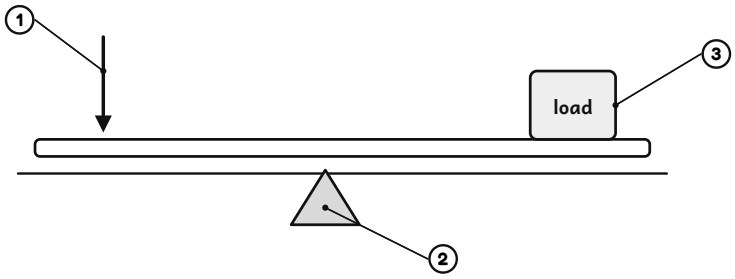


Machine	Letter
wedge	
lever	
screw	
pulley	
inclined plane	
wheel and axle	

2 Label the lever. [3]

LABELS

fulcrum (pivot) **load** **effort (push)**



1 _____ 2 _____ 3 _____

3 **Sort the examples into the simple machine they use.** [9]

SORT THESE

dheki **axe** **bottle opener** **scissors** **door knob** **screwdriver** **rickshaw wheel** **knife** **boti**

Lever	Wedge	Wheel and axle

4 **Fill in the gaps.** [4]

WORD BANK

gravity **friction** **direction** **more** **less** **pivot**

A lever turns on a _____. A pulley can change the _____ of a force. A ramp lets you lift a load with _____ force. Brakes stop a bicycle using _____.

Machines at work

★★☆ Grade 5

Name: Class: Date:

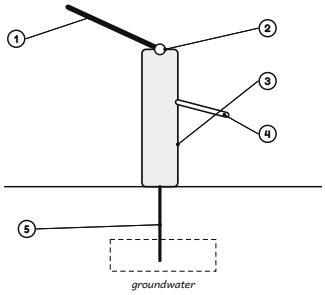
Learning goals

- Explain how simple machines help.
- Analyse a mechanical system.

1 **Rahim pushes a heavy box up a ramp instead of lifting it straight up. What does he gain, and what does he give up?** [2]

2 **Why is it easier to open a door by pushing near the handle than near the hinges?** [2]

3 **Analyse the hand tube-well.** [4]



a) Which simple machine is the handle?

b) Why is the handle long?

c) What force starts the pump, and what is the output?

4 **Name three simple machines on a bicycle or rickshaw and what each does.** [6]

Part	Simple machine	What it does

InquiryMind
LEARN · PLAY · WONDER

How does distance from the pivot change the effort?

★★☆
Grade 5

Name: Class: Date:

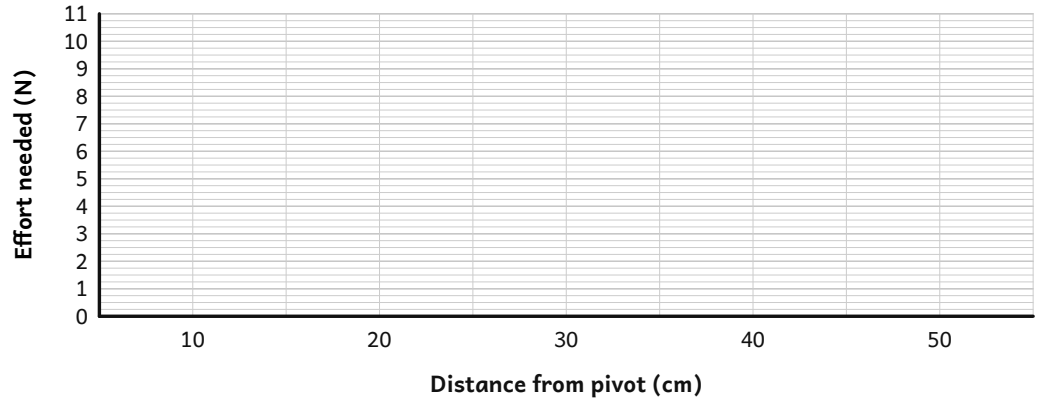
Learning goals

- Measure force with a forcemeter.
- Find a pattern.
- Make a prediction.

Class 5A put a 10 N load on a ruler lever, 10 cm from the pivot. They used a **forcemeter** to measure the effort needed to lift it, pulling at different distances from the pivot on the other side. (N = newton, the unit of force.)

1 **Here are the results. Plot them as a line graph.** [4]

Distance of effort from pivot (cm)	10	20	30	40	50
Effort needed (N)	10	5	3.3	2.5	2



2 **Describe the pattern.** [2]

3 **Predict the effort needed at 60 cm.** [1]

4 **Name two things the class kept the same.** [2]

5 **Use the results to explain why a long crowbar helps to lift a heavy stone.** [1]

REAL-WORLD CASE STUDY · C6 · MECHANICAL SYSTEMS

The rickshaw: a city on three wheels

★★★ Grade 5

Name: Class: Date:

Learning goals

- Analyse a real mechanical system.
- Explain how forces power and stop it.
- Consider people, jobs and technology change.

Science · Design · Economics · Ethics · Art

Dhaka has hundreds of thousands of **cycle rickshaws**, famous for their colourful hand-painted art. A rickshaw is a **mechanical system**: the puller pushes the **pedals** (levers), which turn a **crank** and **gear wheel**. A **chain** carries the turning to a small gear on the back axle, turning the big **wheels**. **Brake levers** press pads against the wheels, and **friction** slows them down.

Pulling a rickshaw is hard work, especially with three passengers or on a bridge. In recent years many rickshaws have been fitted with **batteries and electric motors**. They are faster and easier for the puller, but they can be dangerous at high speed, the batteries must be charged with electricity, and old batteries can pollute if not recycled.

Based on reports about rickshaws and battery rickshaws in Bangladesh.

1 **Complete the table to show how the rickshaw works as a system.** [6]

Part	Simple machine	Job in the system
pedals and crank		
chain and gears		
wheels and axles		
brake levers		

2 **Which force starts the rickshaw moving and which force stops it?** [2]

.....

.....

3 **What might each person think about battery rickshaws?** [3]

Person	View
a rickshaw puller	
a traffic police officer	
an environmental scientist	

4 **Design. Suggest one improvement to make the cycle rickshaw easier to pull or safer. Explain how it uses forces or simple machines.** [2]

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Show what you know

★★★ Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

S	A	X	L	E	V	E	S	B	V
E	H	R	L	P	I	V	O	T	P
A	Y	W	A	L	N	E	L	S	U
G	D	H	F	M	E	E	H	P	L
G	S	E	R	O	P	V	O	V	L
I	C	E	Y	L	R	O	E	V	E
H	R	L	B	S	C	C	H	R	Y
K	E	N	K	N	H	E	E	O	K
N	W	L	V	K	W	E	D	G	E
I	M	K	N	M	F	C	E	S	U

axle force lever pivot pulley ramp screw wedge wheel

2 **Which simple machine is a knife?** [1]

- ◯ pulley
- ◯ screw
- ◯ wedge
- ◯ wheel and axle

3 **What does a single fixed pulley do?** [1]

- ◯ makes the load lighter
- ◯ changes the direction of the force
- ◯ stops friction
- ◯ makes the rope longer

4 **Design a machine to lift a bucket of sand to the top of a wall. Draw and label which simple machines it uses.** [3]

My machine

5 **Challenge. A see-saw has a heavy child on one side and a light child on the other. How can they balance? Explain using the pivot.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
name the six simple machines with examples	○	○	○
label the effort, load and pivot of a lever	○	○	○
explain how simple machines make work easier	○	○	○
find simple machines in a mechanical system	○	○	○
investigate how distance from the pivot changes the effort	○	○	○
explain how forces start and stop a rickshaw	○	○	○